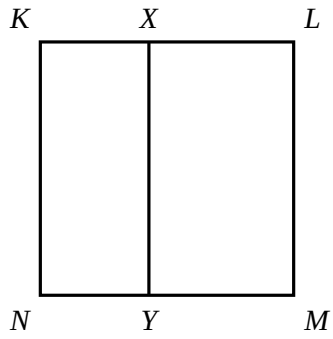
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 7

Part A: Each question is worth 3 points

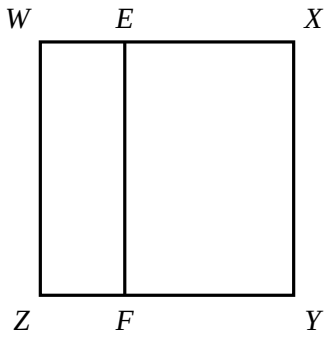
1. *KLMN* is a square. Its side is 14 cm. The line *XY* cuts it into two rectangles, and *KX* = 6 cm.



How many centimetres is the perimeter of the square *KLMN* larger than that of the rectangle *XLMY*?

A. 12 cm ***B.*** 16 cm ***C.*** 24 cm ***D.*** 28 cm ***E.*** 44 cm

2. *WXYZ* is a square. Its side is 18 cm. The line *EF* cuts it into two rectangles, and *WE* = 6 cm.

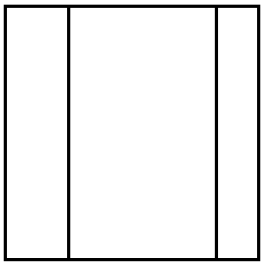


How many centimetres is the perimeter of the square *WXYZ* larger than that of the rectangle *EXYF*?

A. 24 cm ***B.*** 12 cm ***C.*** 11 cm ***D.*** 10 cm ***E.*** 6 cm

Part B: Each question is worth 4 points

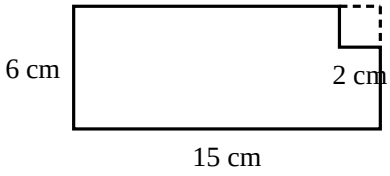
3. A square has perimeter 44 cm. It is cut along straight lines into 3 rectangles as shown.



What is the sum of the perimeters of all 3 rectangles?

A. 132 cm ***B.*** 88 cm ***C.*** 77 cm ***D.*** 66 cm ***E.*** 44 cm

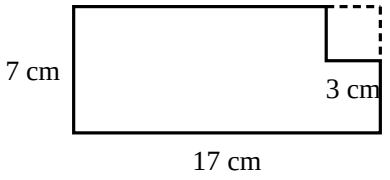
4. A rectangle is 15 cm long and 6 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 42 cm ***B.*** 40 cm ***C.*** 38 cm ***D.*** 34 cm ***E.*** 21 cm

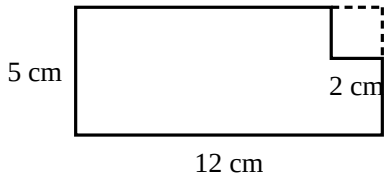
5. A rectangle is 17 cm long and 7 cm wide. A square with side 3 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 48 cm ***B.*** 45 cm ***C.*** 42 cm ***D.*** 36 cm ***E.*** 24 cm

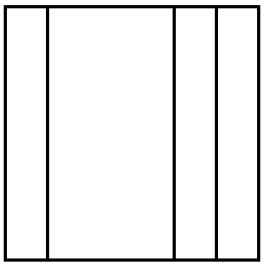
6. A rectangle is 12 cm long and 5 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 17 cm ***B.*** 26 cm ***C.*** 30 cm ***D.*** 32 cm ***E.*** 34 cm

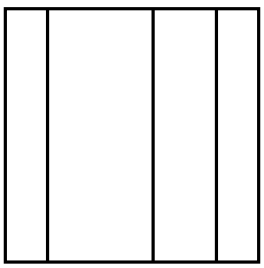
7. A square garden is surrounded by a fence 32 m long. The gardener divides it into 4 rectangular beds as shown and puts a separate fence around each bed.



How many metres of fence are around the beds altogether?

A. 128 m ***B.*** 97 m ***C.*** 80 m ***D.*** 64 m ***E.*** 32 m

8. A square garden is surrounded by a fence 52 m long. The gardener divides it into 4 rectangular beds as shown and puts a separate fence around each bed.

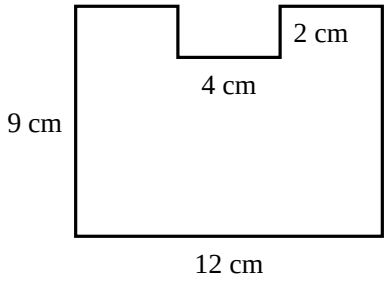


How many metres of fence are around the beds altogether?

A. 208 m ***B.*** 152 m ***C.*** 144 m ***D.*** 130 m ***E.*** 91 m

Part C: Each question is worth 5 points

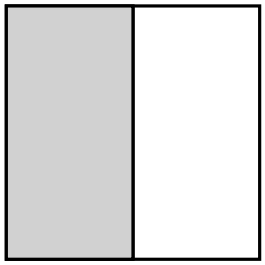
9. A 12 cm by 9 cm rectangle of cardboard has a rectangular notch 4 cm wide and 2 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 38 cm ***B.*** 42 cm ***C.*** 44 cm ***D.*** 45 cm ***E.*** 46 cm

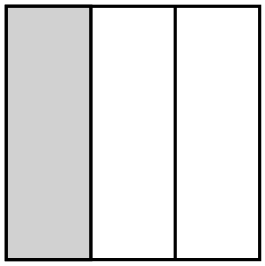
10. Mia cuts a square sheet of paper into 2 equal strips as shown. The perimeter of each strip is 36 cm.



What is the perimeter of the square?

A. 12 cm ***B.*** 48 cm ***C.*** 58 cm ***D.*** 59 cm ***E.*** 72 cm

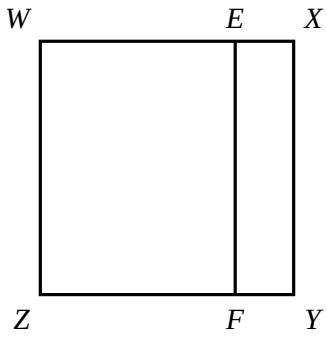
11. Mia cuts a square sheet of paper into 3 equal strips as shown. The perimeter of each strip is 24 cm.



How long is the side of the square?

A. 3 cm ***B.*** 6 cm ***C.*** 7 cm ***D.*** 8 cm ***E.*** 9 cm

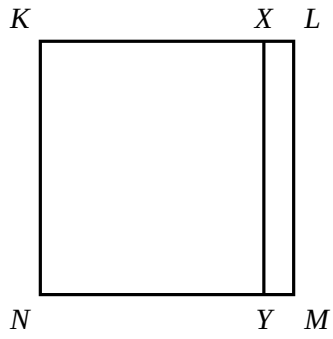
12. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 46 cm and the perimeter of *EXYF* is 32 cm.



What is the perimeter of the square *WXYZ*?

A. 52 cm ***B.*** 26 cm ***C.*** 19 cm ***D.*** 14 cm ***E.*** 13 cm

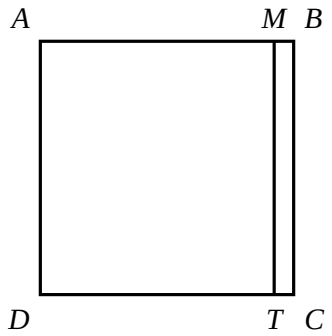
13. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 64 cm and the perimeter of *XLMY* is 38 cm.



How long is the side of the square *KLMN*?

A. 51 cm ***B.*** 34 cm ***C.*** 25 cm ***D.*** 17 cm ***E.*** 13 cm

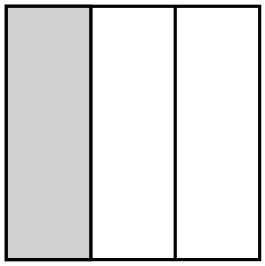
14. The square *ABCD* is cut into two rectangles. The perimeter of *AMTD* is 50 cm and the perimeter of *MBCT* is 28 cm.



What is the perimeter of the square *ABCD*?

A. 22 cm ***B.*** 39 cm ***C.*** 52 cm ***D.*** 65 cm ***E.*** 78 cm

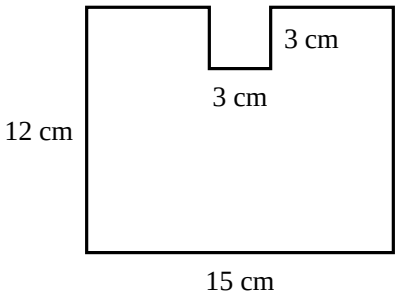
15. Mia cuts a square photo into 3 equal strips as shown. The perimeter of each strip is 40 cm.



How long is the side of the square?

A. 60 cm ***B.*** 25 cm ***C.*** 20 cm ***D.*** 15 cm ***E.*** 5 cm

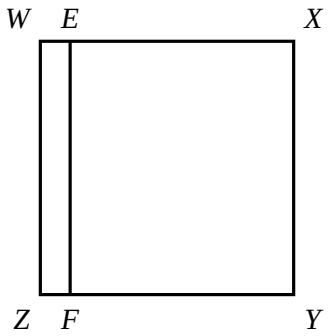
16. A 15 cm by 12 cm rectangle of cardboard has a rectangular notch 3 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 51 cm ***B.*** 60 cm ***C.*** 64 cm ***D.*** 66 cm ***E.*** 71 cm

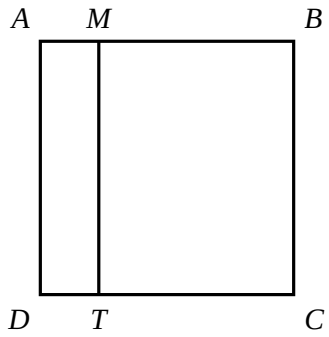
17. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 38 cm and the perimeter of *EXYF* is 64 cm.



How long is the side of the square *WXYZ*?

A. 13 cm ***B.*** 15 cm ***C.*** 16 cm ***D.*** 17 cm ***E.*** 102 cm

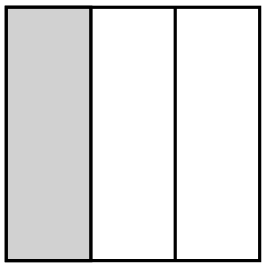
18. The square *ABCD* is cut into two rectangles. The perimeter of *AMTD* is 32 cm and the perimeter of *MBCT* is 46 cm.



What is the perimeter of the square *ABCD*?

A. 13 cm ***B.*** 14 cm ***C.*** 52 cm ***D.*** 65 cm ***E.*** 78 cm

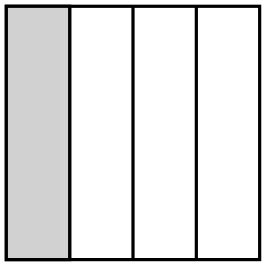
19. Mia cuts a square photo into 3 equal strips as shown. The perimeter of each strip is 48 cm.



What is the perimeter of the square?

A. 18 cm ***B.*** 36 cm ***C.*** 60 cm ***D.*** 65 cm ***E.*** 72 cm

20. Mia cuts a square sheet of paper into 4 equal strips as shown. The perimeter of each strip is 60 cm.



What is the perimeter of the square?

A. 120 cm ***B.*** 96 cm ***C.*** 72 cm ***D.*** 48 cm ***E.*** 24 cm

Answer Key

1.	**A**	**2.**	**B**	**3.**	**B**	**4.**	**A**	**5.**	**A**	**6.**	**E**
7.	**C**	**8.**	**D**	**9.**	**E**	**10.**	**B**	**11.**	**E**	**12.**	**A**
13.	**D**	**14.**	**C**	**15.**	**D**	**16.**	**B**	**17.**	**D**	**18.**	**C**
19.	**E**	**20.**	**B**								

Solutions

1. A (12 cm) — *XLMY* has sides 8 and 14 cm, perimeter 44 cm. Square: 56 cm. Difference 12 cm (= 2 × 6). [2002 Q6 · square_minus_strip]

2. B (12 cm) — *EXYF* has sides 12 and 18 cm, perimeter 60 cm. Square: 72 cm. Difference 12 cm (= 2 × 6). [2002 Q6 · square_minus_strip]

3. B (88 cm) — Side = 44 ÷ 4 = 11 cm. There are 2 cuts; each cut is 11 cm long and is counted twice (once for each piece next to it): 44 + 2 × 2 × 11 = 88 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

4. A (42 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (15 + 6) = 42 cm. [2002 Q6 · corner_cut]

5. A (48 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (17 + 7) = 48 cm. [2002 Q6 · corner_cut]

6. E (34 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (12 + 5) = 34 cm. [2002 Q6 · corner_cut]

7. C (80 m) — Side = 32 ÷ 4 = 8 m. There are 3 cuts; each cut is 8 m long and is counted twice (once for each piece next to it): 32 + 3 × 2 × 8 = 80 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

8. D (130 m) — Side = 52 ÷ 4 = 13 m. There are 3 cuts; each cut is 13 m long and is counted twice (once for each piece next to it): 52 + 3 × 2 × 13 = 130 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

9. E (46 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 2 cm. Perimeter = 2 × (12 + 9) + 2 × 2 = 46 cm. [2002 Q6 · notch]

10. B (48 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 36 ÷ 6 = 6, so the side is 2 × 6 = 12 cm and the perimeter is 48 cm. [2002 Q6 · strips_to_square]

11. E (9 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 24 ÷ 8 = 3, so the side is 3 × 3 = 9 cm. [2002 Q6 · strips_to_square]

12. A (52 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 46 + 32 = 78 cm = 6 sides. Side = 78 ÷ 6 = 13 cm, perimeter 4 × 13 = 52 cm. [2002 Q6 · two_perimeters]

13. D (17 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 64 + 38 = 102 cm = 6 sides. Side = 102 ÷ 6 = 17 cm. [2002 Q6 · two_perimeters]

14. C (52 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 50 + 28 = 78 cm = 6 sides. Side = 78 ÷ 6 = 13 cm, perimeter 4 × 13 = 52 cm. [2002 Q6 · two_perimeters]

15. D (15 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 40 ÷ 8 = 5, so the side is 3 × 5 = 15 cm. [2002 Q6 · strips_to_square]

16. B (60 cm) — The bottom of the notch (3 cm) replaces the 3 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (15 + 12) + 2 × 3 = 60 cm. [2002 Q6 · notch]

17. D (17 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 38 + 64 = 102 cm = 6 sides. Side = 102 ÷ 6 = 17 cm. [2002 Q6 · two_perimeters]

18. C (52 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 32 + 46 = 78 cm = 6 sides. Side = 78 ÷ 6 = 13 cm, perimeter 4 × 13 = 52 cm. [2002 Q6 · two_perimeters]

19. E (72 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 48 ÷ 8 = 6, so the side is 3 × 6 = 18 cm and the perimeter is 72 cm. [2002 Q6 · strips_to_square]

20. B (96 cm) — A strip is the full side long and 1/4 of it wide, so its perimeter is side + side + 2 × (side ÷ 4) = 10 × (side ÷ 4). 60 ÷ 10 = 6, so the side is 4 × 6 = 24 cm and the perimeter is 96 cm. [2002 Q6 · strips_to_square]